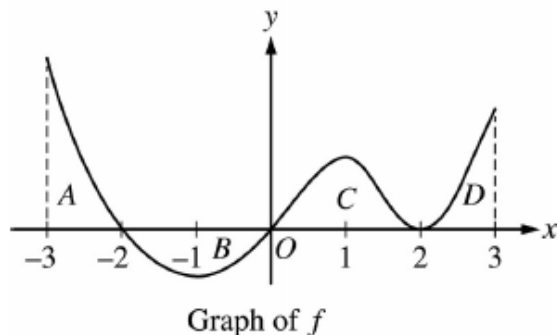


“Speaking of leeching off of your friends, don’t do it. It is easy to take advantage of the freedom that Mr. Gough allows you in completing assignments. At the end of the year, it becomes very clear as to who did the work and who did not. The problem sets are an excellent tool in preparing for the exam, so do them.” A 2019 Nerd

FR1. Practice Test 2016

No calculator is allowed for problems on this part of the exam.



The graph of a differentiable function f is shown above for $-3 \leq x \leq 3$. The graph of f has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 2$. The areas of regions A , B , C , and D are 5, 4, 5, and 3, respectively. Let g be the antiderivative of f such that $g(3) = 7$.

- Find all values of x on the open interval $-3 < x < 3$ for which the function g has a relative maximum. Justify your answer.
- On what open intervals contained in $-3 < x < 3$ is the graph of g concave up? Give a reason for your answer.
- Find the value of $g(0)$.
- Find the value of $\lim_{x \rightarrow 0} \frac{g(x)+1}{2x}$, or state that it does not exist. Show the work that leads to your answer.

FR2. 2005B BC 1 (Calculator)

An object moving along a curve in the xy -plane is at position $(x(t), y(t))$ at time $t \geq 0$, with

$$\frac{dx}{dt} = 18t - 6t^2 \text{ and } \frac{dy}{dt} = \ln(1 + (t-4)^4).$$

At time $t = 0$, the object is at position $(-13, 5)$. At time $t = 2$, the object is at point P with x -coordinate 7.

- Find the acceleration vector at time $t = 2$ and the speed of the object at time $t = 2$.
- Find the y -coordinate of P .
- Write an equation for the line tangent to the curve at P .
- For what value of t , if any, is the object at rest? Explain your reasoning.

MC1. Let g be a continuous function on the closed interval $[0,1]$. Let $g(0) = 1$ and $g(1) = 0$. Which of the following is NOT necessarily true?

- (A) There exists a number h in $[0,1]$ such that $g(h) \geq g(x)$ for all x in $[0,1]$.
- (B) For all a and b in $[0,1]$, if $a = b$, then $g(a) = g(b)$.
- (C) There exists a number h in $[0,1]$ such that $g(h) = \frac{1}{2}$.
- (D) There exists a number h in $[0,1]$ such that $g(h) = \frac{3}{2}$.

MC2. If $x = t^3 - t$ and $y = \sqrt{3t+1}$, then $\frac{dy}{dx}$ at $t = 1$ is

- (A) $\frac{1}{8}$ (B) $\frac{3}{8}$ (C) $\frac{3}{4}$ (D) $\frac{8}{3}$

MC3. CA At time $t \geq 0$, a particle moving in the xy -plane has velocity vector given by $\mathbf{v}(t) = \langle 4e^{-t}, \sin(1 + \sqrt{t}) \rangle$. What is the total distance the particle travels between $t = 1$ and $t = 3$?

- (A) 1.861
- (B) 1.983
- (C) 2.236
- (D) 4.851

MC4. The position of a particle moving in the xy -plane is given by the parametric equations $x(t) = \frac{6t}{t+1}$ and $y(t) = \frac{-8}{t^2+4}$. What is the slope of the line tangent to the path of the particle at the point where $t = 2$?

- (A) $\frac{1}{2}$
- (B) $\frac{2}{3}$
- (C) $\frac{3}{4}$
- (D) $\frac{4}{3}$